"Least" action principle in relativity

$L = \frac{1}{2} m \dot{\vec{r}}^2 - U(\vec{r})$ is not correct Lagrangian for relativistic system.

1) K.E. is not correct

2) The concept of potential energy is not valid — interactions between particles propagate with finite speed ($\leq c$) and therefore interactions must also depend on time.

We seek action that is invariant under the change of reference frame

$\Rightarrow$ S is 4-scalar. Consider free particle moving along its world line: $dx^\alpha u_\alpha = dx^\alpha \frac{dx_\alpha}{d\tau} = \frac{ds^2}{d\tau} = c^2 d\tau = c\, ds$

4-velocity

$$ds^2 = c^2 dt^2 - d\vec{r} \cdot d\vec{r} = c^2 d\tau^2$$

proper time

$$S_{free} = Kc^2 \int_{(ct_1, \vec{r}_1)}^{(ct_2, \vec{r}_2)} d\tau = Kc \int_1^2 ds = \int_1^2 dt\, Kc^2 \frac{d\tau}{dt}$$

some constant

$$L = Kc^2 \frac{d\tau}{dt} = Kc^2 \sqrt{1 - v^2/c^2}$$

For non-relativistic motion

$$L \cong Kc^2 \left(1 - \frac{v^2}{2c^2}\right) = Kc^2 - \frac{1}{2} K v^2$$

matches $\frac{1}{2} m v^2$ if $K = -m$. $\Rightarrow L = -mc^2 \sqrt{1 - v^2/c^2}$

$$\Rightarrow S_{free} = -mc^2 \int_1^2 d\tau$$

Why the <u>minus</u> sign? Consider particle that starts and ends its motion at the same space point, $\vec{r}_1 = \vec{r}_2$.

For particle that does not move, the interval is just

$$s = c\int dt = c\Delta t .$$

For particle which moves $s' = \int\sqrt{c^2dt^2 - (d\vec{r})^2} = c\int dt\sqrt{1-\frac{v^2}{c^2}}$

But $\sqrt{1-\frac{v^2}{c^2}} < 1 \Rightarrow s' < s.$

Therefore the optimal trajectory (with s) corresponds to the interval which is <u>maximum</u>. To make action <u>minimal</u> we need $S = -mc\int ds$.

Momentum $\vec{p} = \frac{\partial L}{\partial \vec{v}} = -mc^2 \frac{\partial}{\partial \vec{v}}\sqrt{1-v^2/c^2} =$

$$= -mc^2 \frac{-2\vec{v}/c^2}{2\sqrt{1-v^2/c^2}} = \frac{m\vec{v}}{\sqrt{1-v^2/c^2}}$$

not components of 4-momentum yet.

$$\mathcal{E} = \vec{v}\cdot\frac{\partial L}{\partial \vec{v}} - L = \frac{mv^2}{\sqrt{1-v^2/c^2}} + mc^2\sqrt{1-v^2/c^2}$$

$$= \frac{1}{\sqrt{1-v^2/c^2}}\left[mv^2 + mc^2\left(1-\frac{v^2}{c^2}\right)\right] = \frac{mc^2}{\sqrt{1-v^2/c^2}}$$

EL eq: $\underbrace{\frac{\partial L}{\partial \vec{r}}}_{=0} - \frac{d}{dt}\frac{\partial L}{\partial \vec{v}} = 0 \Rightarrow \frac{d}{dt}\frac{m\vec{v}}{\sqrt{1-v^2/c^2}} = 0.$

$\Rightarrow \dot{\vec{v}} = 0$

$$\Rightarrow S_{free} = -mc\int_{1}^{2} ds = -mc\,s = -mc^2\underbrace{\sqrt{1-v^2/c^2}\,\Delta t}_{\Delta\tau}$$

$$S = c\Delta\tau = \sqrt{(x^\alpha_{(2)} - x^\alpha_{(1)})(x_{(2)\alpha} - x_{(1)\alpha})}$$

let for simplicity $x_1^{(\alpha)} = 0$ so that $s = \sqrt{x^\alpha_{(2)} x_{(2)\alpha}} = \sqrt{x^\alpha x_\alpha}$

initial point

end-point of particles trajectory

$S = -mcs$:

$$-\frac{\partial S}{\partial x_\beta} = mc\frac{\partial s}{\partial x_\beta} = mc\frac{\frac{\partial x_\alpha}{\partial x_\beta}x^\alpha + \frac{\partial x^\alpha}{\partial x_\beta}x_\alpha}{2\sqrt{x^\mu x_\mu}}$$

$$= mc\frac{2\frac{dx_\alpha}{dx_\beta}x^\alpha}{2\sqrt{x^\mu x_\mu}} = mc\frac{\delta^\beta_\alpha x^\alpha}{c\Delta\tau} = m\frac{x^\beta}{\Delta\tau} = mu^\beta = p^\beta$$

4-velocity 4-momentum

$$u^\beta = \gamma(c, \vec{v})$$

$$p^\beta = m\gamma(c, \vec{v}) = \left(\frac{\varepsilon}{c}, m\gamma\vec{v}\right) = \left(\frac{\varepsilon}{c}, \vec{p}\right).$$

$p^\alpha = -\frac{\partial S}{\partial x_\alpha}$ where x_α is the end-point of the <u>optimal</u> trajectory.

So far this is all about 1 free particle.

<u>4-calculus</u> scalar ϕ

$$\frac{\partial\phi}{\partial x^\alpha} = \left(\frac{1}{c}\frac{\partial\phi}{\partial t}, \vec{\nabla}\phi\right), \quad d\phi = \frac{\partial\phi}{\partial x^i}dx^i \Rightarrow g_i = \frac{\partial\phi}{\partial x^i} \text{ is}$$

scalar

covariant vector.

$$\frac{\partial\phi}{\partial x_\alpha} = \left(\frac{1}{c}\frac{\partial\phi}{\partial t}, -\vec{\nabla}\phi\right), \quad d\phi = \frac{\partial\phi}{\partial x_i}dx_i = g^i dx_i$$

$$\Rightarrow \frac{\partial\phi}{\partial x_i} = g^i \text{ is contravariant.}$$

Collisions

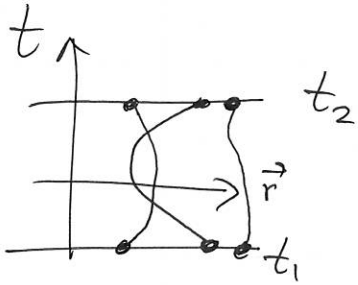
$$S = S_{free} + S_{col}$$

$$-c \sum_i m_i \int_{X_i^\alpha}^{x_i^\alpha} ds_i$$

x_i^α ← final

X_i^α ← initial

action due to interaction between particles when they are close to each other. Takes place within <u>small</u> area of space-time



Total S is dominated by S_{free} for times long before and long after the collision.

Let us now translate all world lines by a fixed 4-vector $\Delta\xi^\alpha$, including the boundary points

$$X_i^\alpha \to X_i^\alpha + \Delta\xi^\alpha, \quad x_i^\alpha \to x_i^\alpha + \Delta\xi^\alpha.$$

From homogeneity of space-time $\Delta S = 0$!

$$\Delta S_{col} = 0.$$

$$\Rightarrow \Delta S = \sum_i \left(\underbrace{\frac{\partial S_{free}}{\partial x_i^\alpha}}_{= -p_{i\alpha} \text{ initial}} \Delta\xi^\alpha + \underbrace{\frac{\partial S_{free}}{\partial X_i^\alpha}}_{= P_{i\alpha} \text{ final}} \Delta\xi^\alpha \right)$$

$$\Rightarrow \Delta S = \Delta\xi^\alpha \sum_i \left(-p_{i\alpha} + P_{i\alpha} \right) = 0$$

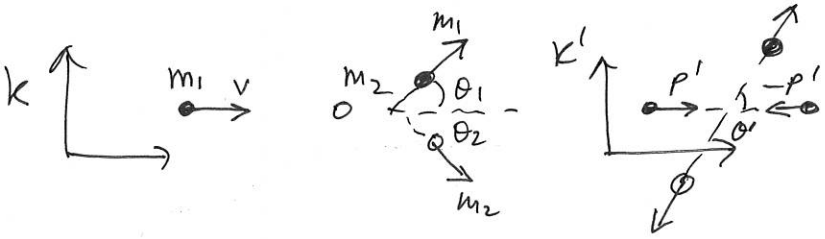
$$\Rightarrow \sum_i P_{i\alpha} = \sum_i p_{i\alpha}$$ 4-momentum conservation.

Elastic collision of 2 particles.

$$\begin{pmatrix} \mathcal{E}'/c \\ p'_x \end{pmatrix} = \gamma \begin{pmatrix} 1 & -\beta \\ -\beta & 1 \end{pmatrix} \begin{pmatrix} \mathcal{E}/c \\ p_x \end{pmatrix} \Rightarrow \begin{array}{l} \mathcal{E}' = \gamma\left(\mathcal{E} - \frac{V}{c} p_x c\right) \\ p'_x = \gamma\left(p_x - \frac{\beta}{c}\mathcal{E}\right) \end{array}$$

$$\Rightarrow \sum p'_x = \gamma\left(\sum p_x - \frac{V}{c^2}\sum \mathcal{E}\right)$$

In the CM frame $\sum p'_x = 0 \Rightarrow V_{CM} = \dfrac{c^2 \sum p_x}{\sum \mathcal{E}}$.



$$\left.\begin{array}{ll} \mathcal{E}_1 = \dfrac{m_1 c^2}{\sqrt{1-v^2/c^2}} & p_1 = \dfrac{mv}{\sqrt{1-v^2/c^2}} \\ \mathcal{E}_2 = m_2 c^2 & p_2 = 0 \end{array}\right\} \quad V_{CM} = \frac{c^2 \dfrac{m_1 v}{\sqrt{1-v^2/c^2}}}{\dfrac{m_1 c^2}{\sqrt{1-v^2/c^2}} + m_2 c^2} = \frac{m_1 v\sqrt{1-v^2/c^2}}{m_1 + m_2\sqrt{1-v^2/c^2}}$$

In CM frame = K' frame : total momentum = 0

$\Rightarrow$ particles have equal and opposite $\vec{p}_i{}'$ and $-\vec{p}_i{}'$.

$\Rightarrow$ after the collision $\vec{p}_f{}'$ and $-\vec{p}_f{}'$

For every particle $p^i p_i = m^2 c^2$

$$\mathcal{E}^2 = m^2 c^4 + c^2 \vec{p}^{\,2}$$

$\mathcal{E}$ is conserved $\Rightarrow |\vec{p}_i{}'| = |\vec{p}_f{}'|$.

$\Rightarrow$ the only degree of freedom is angle θ'.

Find incident momentum $\vec{p}_i'$: $\quad -p_i' = \gamma\left(\underset{0}{\underset{\shortparallel}{p_2}} - \frac{\beta}{c}\underset{m_2c^2}{\underset{\shortparallel}{\varepsilon_2}}\right)$

$$p_i' = \gamma\beta c\, m_2 = m_2\gamma V_{cm}.$$

After scattering

particle 1

$$p_{1x}' = P_f \cos\theta' = m_2\gamma V_{cm}\cos\theta'$$

$$p_{1y}' = P_f \sin\theta' = m_2\gamma V_{cm}\sin\theta'$$

$$\frac{\varepsilon_1'}{c} = \sqrt{m_1^2c^2 + p'^2} = \sqrt{m_1^2c^2 + m_2^2\gamma^2 V_{cm}^2} = c\sqrt{m_1^2 + \gamma^2\beta^2 m_2^2}$$

$$\beta = \frac{V_{cm}}{c}$$

$$\frac{\varepsilon_2'}{c} = \sqrt{m_2^2c^2 + p'^2} = c\sqrt{m_2^2 + \gamma^2\beta^2 m_2^2} = \gamma m_2 c.$$

$$\begin{pmatrix} \frac{\varepsilon_1}{c} \\ p_{1x} \end{pmatrix} = \gamma\begin{pmatrix} 1 & \beta \\ \beta & 1 \end{pmatrix}\begin{pmatrix} \varepsilon_1'/c \\ p_{1x}' \end{pmatrix} \quad \text{and} \quad p_{1y} = p_{1y}'$$

$$p_{1x} = \gamma\left(\beta\frac{\varepsilon_1'}{c} + p_{1x}'\right) = \gamma\left(V_{cm}\sqrt{m_1^2 + \gamma^2\beta^2 m_2^2} + m_2\gamma V_{cm}\cdot\cos\theta'\right)$$

$$\Rightarrow \tan\theta_1 = \frac{p_{1y}}{p_{1x}} = \frac{\gamma V_{cm}\, m_2 \sin\theta'}{\gamma V_{cm}\left(m_2\gamma\cos\theta' + \sqrt{m_1^2 + \gamma^2\beta^2 m_2^2}\right)}$$

$$= \frac{m_2 \sin\theta'}{m_2\gamma\cos\theta' + \sqrt{m_1^2 + \gamma^2\beta^2 m_2^2}}$$